

HDOC – Day 31

C Programming Assignment

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Problem Statement

Write a C program to perform the following menu-driven operations on a given 2D array: (1) find the sum of all elements, (2) find the sum of each row and store it in an array, (3) find the transpose of the matrix, and (4) exit.

Algorithm

1. Start the program.
2. Read the number of rows and columns.
3. Read all matrix elements into a 2D array.
4. Repeatedly display/read a menu choice until the user chooses Exit.
5. For choice 1, initialize total = 0 and add every matrix element to total; display total.
6. For choice 2, for each row initialize its sum to 0, add all elements of that row, store the result in a row-sum array, and display the array.
7. For choice 3, print matrix[j][i] for each column i and row j. This produces a matrix of size columns x rows.
8. For choice 4, terminate the menu loop.
9. For any other choice, display an invalid-choice message.
10. Stop.

C Program

```
#include <stdio.h>
#define MAX 100
int main(void)
{
    int matrix[MAX][MAX], rowSum[MAX];
```

```

int rows, cols, i, j, choice;
printf("Enter the number of rows and columns: ");
scanf("%d %d", &rows, &cols);
if (rows <= 0 || cols <= 0 || rows > MAX || cols > MAX) {
printf("Invalid matrix size.\n");
return 0;
}
printf("Enter the matrix elements:\n");
for (i = 0; i < rows; i++) {
for (j = 0; j < cols; j++) {
scanf("%d", &matrix[i][j]);
}
}
do {
printf("\n1. Sum of all elements\n");
printf("2. Sum of each row\n");
printf("3. Transpose of matrix\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1: {

int total = 0;
for (i = 0; i < rows; i++)
for (j = 0; j < cols; j++)
total += matrix[i][j];
printf("Sum of all elements = %d\n", total);
break;
}
case 2:
printf("Row sums:\n");
for (i = 0; i < rows; i++) {
rowSum[i] = 0;
for (j = 0; j < cols; j++)

```

```

rowSum[i] += matrix[i][j];
printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
}
printf("The row-sum array contains: [");
for (i = 0; i < rows; i++) {
printf("%d", rowSum[i]);
if (i < rows - 1)
printf(", ");
}
printf("]\n");
break;
case 3:
printf("Transpose of the matrix:\n");
for (j = 0; j < cols; j++) {
for (i = 0; i < rows; i++)
printf("%d ", matrix[i][j]);
printf("\n");
}
break;
case 4:
printf("Exiting program.\n");
break;
default:
printf("Invalid choice. Please enter 1 to 4.\n");
}
} while (choice != 4);
return 0;
}

```

Sample Execution

Enter the number of rows and columns: 2 3

Enter the matrix elements:

1 2 3

4 5 6

1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit

Enter your choice: 1

Sum of all elements = 21

Enter your choice: 2

Row sums:

Sum of row 1 = 6

Sum of row 2 = 15

The row-sum array contains: [6, 15]

Enter your choice: 3

Transpose of the matrix:

1 4

2 5

3 6

Enter your choice: 4

Exiting program.

Result

The program successfully performs all the required menu-driven operations on a 2D array: total sum, row-wise sums stored in an array, transpose, and exit.